Year 10 Mathematics

Shape and Space 1

5.2 Quadrilaterals

Quadrilaterals

STARTER · 5 MIN

Date: _____ **Title:** Quadrilaterals

LEARNING OBJECTIVES

- Use the angle sum of a quadrilateral
- Use the properties of special quadrilaterals
- Solve angle problems in quadrilaterals

STARTER *Answer in your book. No calculators.*

1. Work out 360 − 250
2. Work out 180 − 75
3. Work out 360 − 90 − 90 − 120
4. Solve 4x = 360

Starter — Answers

STARTER

Tick or correct in green pen

1. $360 - 250 =$ **110**
2. $180 - 75 =$ **105**
3. $360 - 300 =$ **60**
4. $x = 360 \div 4 =$ **90**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use the angle sum of a quadrilateral	☐
SILVER	I can use the properties of special quadrilaterals	☐
GOLD	I can solve angle problems in quadrilaterals	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Quadrilaterals

INTRODUCTION · 3 MIN

The four angles in any quadrilateral add up to 360°.

Angle sum

a + b + c + d = 360°

85° + 95° + 110° + 70° = 360°

Parallelogram

Opposite angles are equal.

Next to each other add to 180°

Kite

One pair of opposite angles is equal.

Two pairs of equal sides

Trapezium

One pair of parallel sides.

Angles between them add to 180°

Examples 1 and 2 — Angle Sum of a Quadrilateral

EXAMPLES · 5 MIN

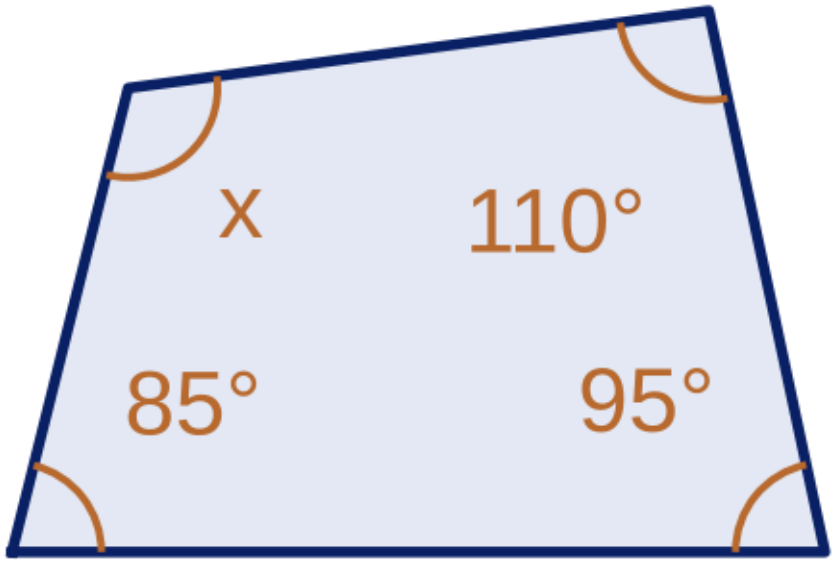
BRONZE

Watch and copy into your book

Hint a + b + c + d = 360°

Example 1

Find the size of angle x.



Example 2

ABCD is a parallelogram.

Angle A = 65°.

Find angles B, C and D.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$x = 360 - 85 - 95 - 110$

$x = 70°$

Tip: angles in a quadrilateral add up to 360°.

Example 2

$C = A$

$= 65°$

$B = 180 - 65$

$= 115°$

$D = B$

$= 115°$

Tip: opposite angles in a parallelogram are equal.

Task 1

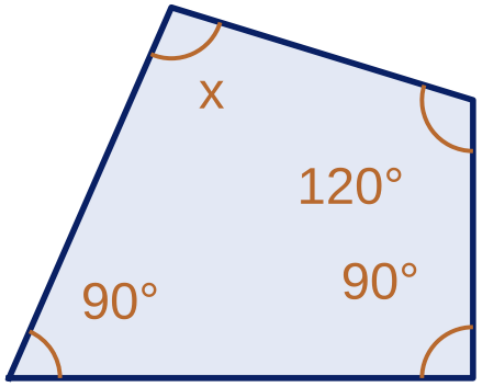
PRACTICE · 9 MIN

BRONZE

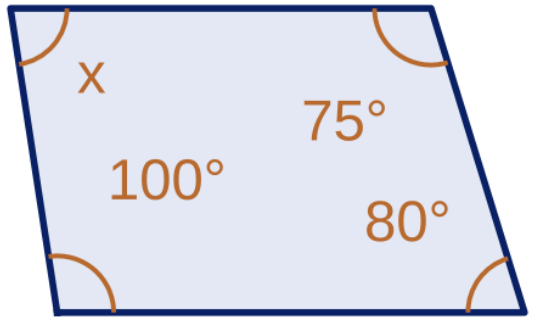
Find the size of each angle marked x

Hint a + b + c + d = 360°

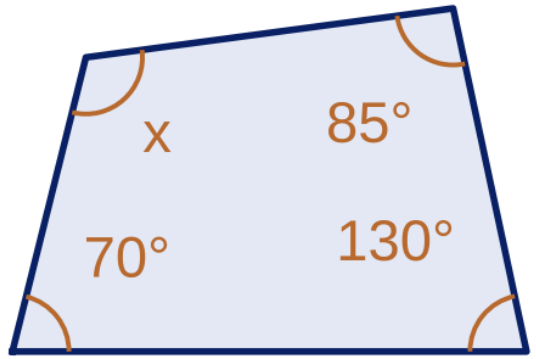
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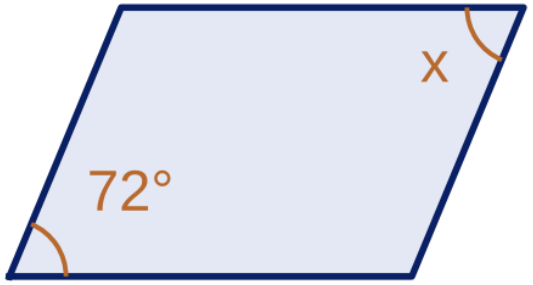
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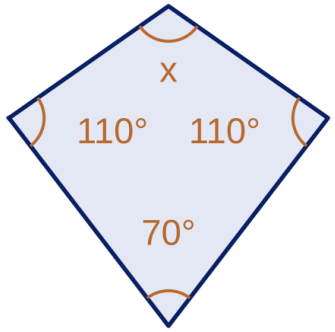
3



4 Parallelogram



5 Kite



★ Write down the size of each angle in a rectangle.

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $x = 360 - 90 - 90 - 120 =$ **60°**
2. $x = 360 - 100 - 80 - 75 =$ **105°**
3. $x = 360 - 70 - 130 - 85 =$ **75°**
4. $x =$ **72°**
5. $x = 360 - 110 - 110 - 70 =$ **70°**

★ Each angle = **90°**

Examples 3 and 4 — Special Quadrilaterals

EXAMPLES · 5 MIN

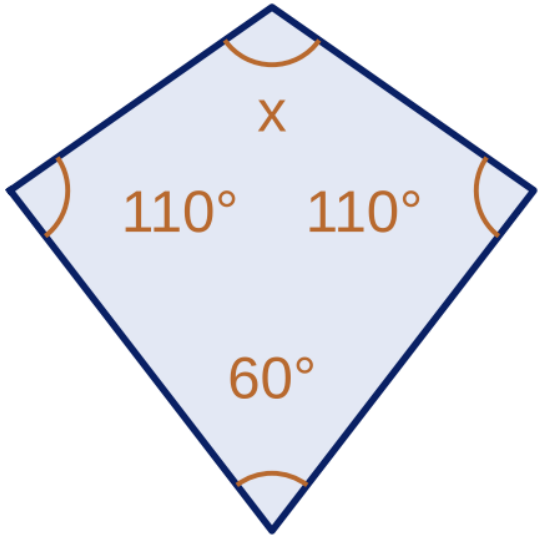
SILVER

Watch and copy into your book

Hint a + b + c + d = 360° co-interior angles add up to 180°

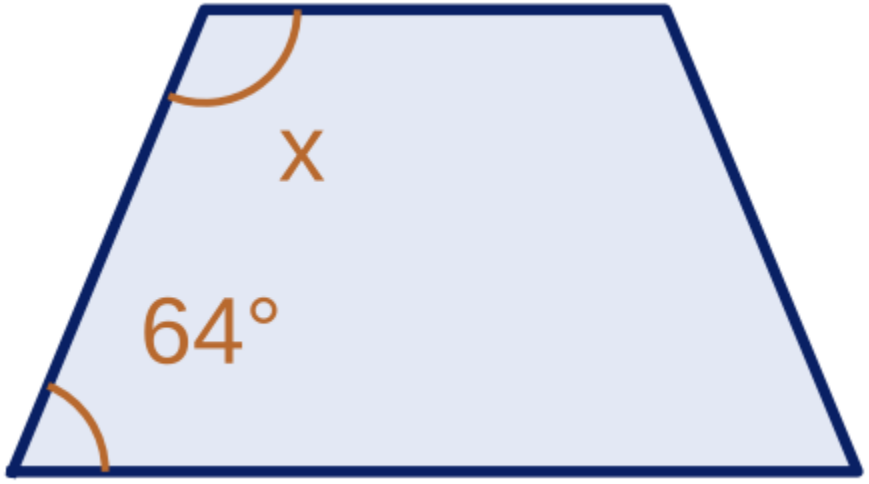
Example 3

The kite has two equal angles of 110°. Find x.



Example 4

ABCD is a trapezium. Find angle x.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$x = 360 - 110 - 110 - 60$

$x = 80°$

Tip: the equal angles are between the unequal sides.

Example 4

$x = 180 - 64$

$x = 116°$

Tip: angles between parallel sides add to 180°.

Task 2

PRACTICE · 9 MIN

SILVER

Answer in your book

Hint a + b + c + d = 360° co-interior angles add up to 180°

1. A kite has equal angles of 95°. Its third angle is 70°. Find the fourth angle.
2. A trapezium has base angles of 48° and 75°. Find the other two angles.
3. One angle of a rhombus is 50°. Find the other three angles.

4. Name the quadrilateral: four equal sides, diagonals cross at 90°, all angles 90°.
5. Name the quadrilateral with exactly one pair of parallel sides.

Task 2 — Answers

CORRECTIONS

SILVER *Tick or correct in green pen*

1. $360 - 95 - 95 - 70 =$ **100°**

2. $180 - 48 = 132°$, $180 - 75 =$ **105°**

3. $180 - 50 = 130°$, **Angles: 50°, 130°, 130°**

4. **Square**

5. **Trapezium**

Examples 5 and 6 — Quadrilateral Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint
a + b + c + d = 360°
co-interior angles add up to 180°

Example 5

(3 marks)

The angles of a quadrilateral are

x, 2x, 3x and 4x.

Find the largest angle.

Example 6

(3 marks)

In a parallelogram, two angles next to each other

are 3x + 10 and 2x + 20.

Find both angles.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$10x = 360$$

$$x = 36$$

$$\text{Largest} = 4 \times 36$$

$$= 144°$$

Tip: collect the x terms first.

Example 6

$$3x + 10 + 2x + 20 = 180$$

$$5x = 150$$

$$x = 30$$

Angles: 100° and 80°

Tip: angles next to each other add to 180°.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint
a + b + c + d = 360°
co-interior angles add up to 180°

1. The angles of a quadrilateral are x, x + 20, 2x and 2x – 20. Find x. (3 marks)

2. Opposite angles of a parallelogram are 5x – 20 and 3x + 40. Find the angle. (3 marks)

3. A kite has angles 2x, 2x, x + 20 and x + 40. Find x. (3 marks)

4. In a trapezium, two angles between the parallel sides are 4x and 2x + 30. Find x. (3 marks)

5. The angles of a quadrilateral are in the ratio 2 : 3 : 3 : 4. Find the largest angle. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $6x = 360, \quad x = \mathbf{60°}$

2. $5x - 20 = 3x + 40, \quad x = 30, \quad \text{Angle} = \mathbf{130°}$

3. $6x + 60 = 360, \quad 6x = 300, \quad x = \mathbf{50°}$

4. $6x + 30 = 180, \quad 6x = 150, \quad x = \mathbf{25°}$

5. $360 \div 12 = 30, \quad 4 \times 30 = \mathbf{120°}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Three angles of a quadrilateral are 95°, 85° and 110°. Find the fourth.

SILVER One angle of a parallelogram is 58°. Find the angle next to it.

GOLD The angles of a quadrilateral are x, 2x, x + 40 and 2x + 20. Find x.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $360 - 95 - 85 - 110 =$ **70°**

SILVER $180 - 58 =$ **122°**

GOLD $6x + 60 = 360, \quad 6x = 300, \quad x =$ **50°**

Exam Practice — Angles in Quadrilaterals

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

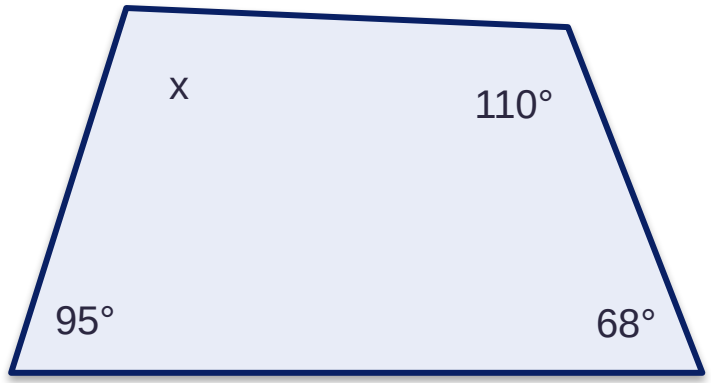
Hint a + b + c + d = 360°

Question 1

(2 marks)

Work out the size of angle x.

Diagram NOT accurately drawn



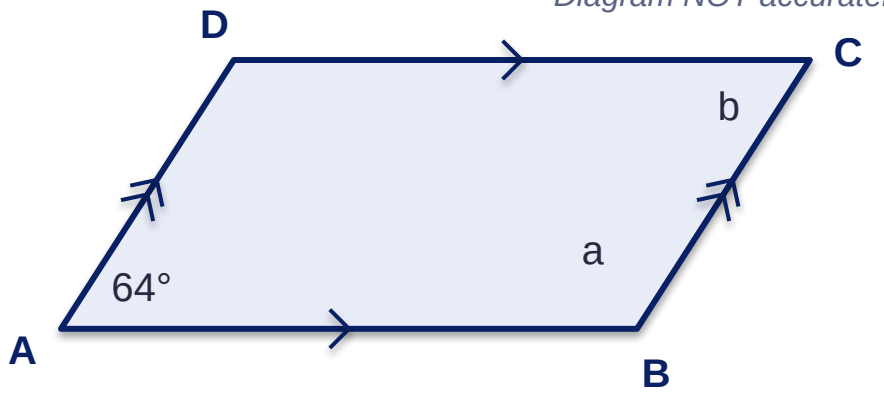
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Question 2

(2 marks)

ABCD is a parallelogram.

Diagram NOT accurately drawn



(a) Work out the size of angle a. (1)

..... °

(b) Work out the size of angle b. (1)

..... °

Exam Practice — Kites and Trapeziums

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint
a + b + c + d = 360°
co-interior angles add up to 180°

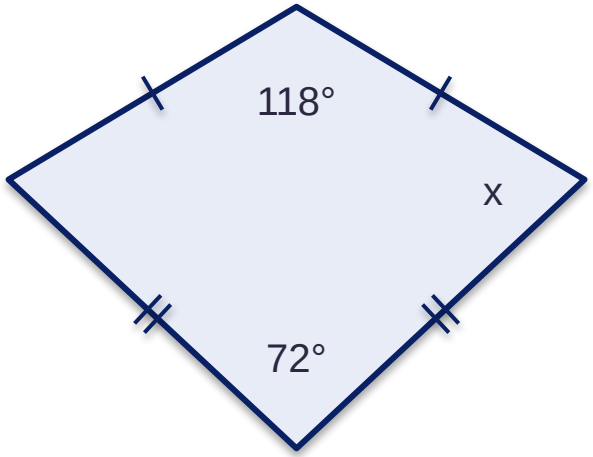
Question 3

(3 marks)

The diagram shows a kite.

Work out the size of angle x.

Diagram NOT accurately drawn



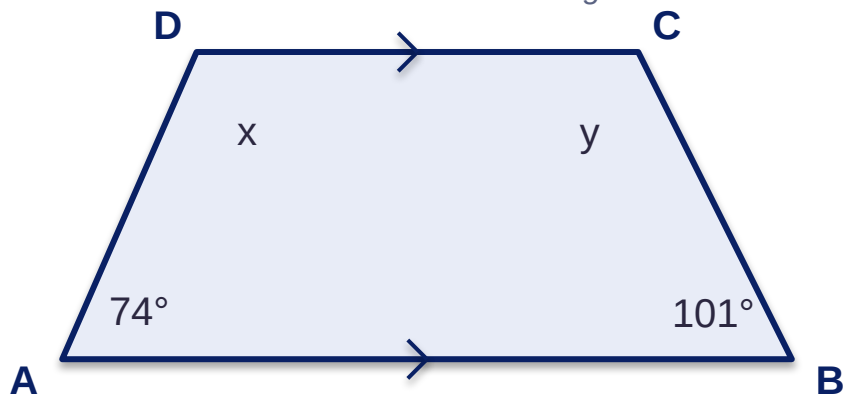
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Question 4

(3 marks)

ABCD is a trapezium.

Diagram NOT accurately drawn



(a) Work out the size of angle x. (1)

..... °

(b) Work out the size of angle y. (2)

..... °

Exam Practice — Algebra in Quadrilaterals

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint
a + b + c + d = 360°
co-interior angles add up to 180°

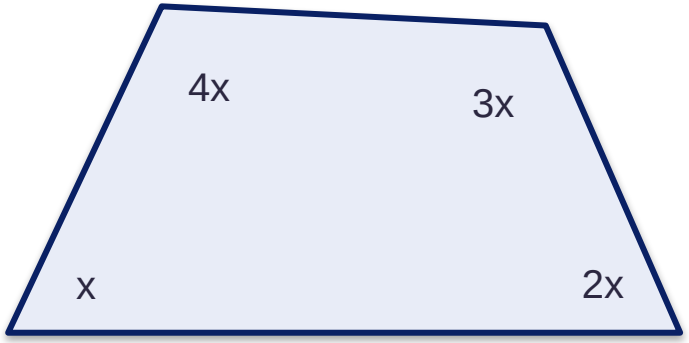
Question 5

(3 marks)

The angles of a quadrilateral are x, 2x, 3x and 4x.

Work out the size of the largest angle.

Diagram NOT accurately drawn



..... °

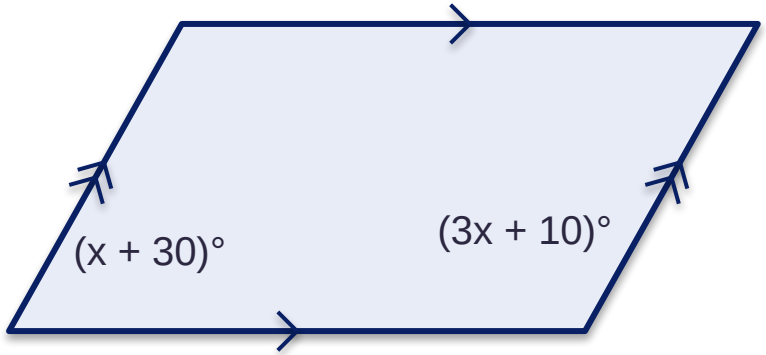
Question 6

(3 marks)

ABCD is a parallelogram.

Work out the size of the smallest angle.

Diagram NOT accurately drawn



..... °

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

360 − 95 − 68 − 110 — M1

x = 87° — A1

Tip: angles in a quadrilateral add to 360°.

Question 2

(a) a = 180 − 64 = 116° — B1

(b) b = 64° — B1

Tip: opposite angles equal; co-interior sum 180°.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

360 − 118 − 72 = 170 — M1

170 ÷ 2 — M1

x = 85° — A1

Tip: a kite has one pair of equal angles.

Question 4

(a) x = 180 − 74 = 106° — B1

(b) 180 − 101 — M1

y = 79° — A1

Tip: co-interior angles add to 180°.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$10x = 360$	M1
$x = 36$	M1
$4x = 144°$	A1

Tip: x + 2x + 3x + 4x = 10x.

Question 6

$x + 30 + 3x + 10 = 180$	M1
$4x = 140,\ \ x = 35$	M1
Smallest = 65°	A1

Tip: adjacent angles in a parallelogram add to 180°.